# CHAPTER 2: PROBABILITY AND COUNTING RULES

PART D

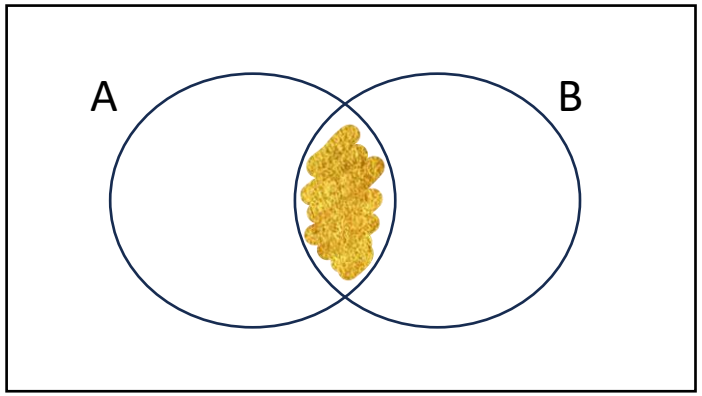
- CONDITIONAL PROBABILITY
- INDEPENDENT EVENTS
- PRODUCT RULES OR MULTIPLICATIVE RULES
- BAYES RULES

# Conditional Probability

- **Definition 6:** The probability of an event *B* occurring when it is known that some event *A* has occurred (probability of event *B*, happening, given that *A* has happened) is called a conditional probability and is denoted by $P(B|A)$**,** and is defined by

$$P(B|A) = \frac{P(B \cap A)}{P(A)}$$

# Example 28




*A* and *B* are two events with $P(A) = \frac{1}{3}, P(B) = \frac{1}{2}$ and $P(A \cup B) = \frac{3}{4}$ ,find:

a) $P(B|A)$

b) $P(A'|B')$

c) $P(A'|B)$

a) $P(B|A) = \frac{P(B \cap A)}{P(A)}$

$= \frac{{}^{1}/_{12}}{{}^{1}/_{3}}$

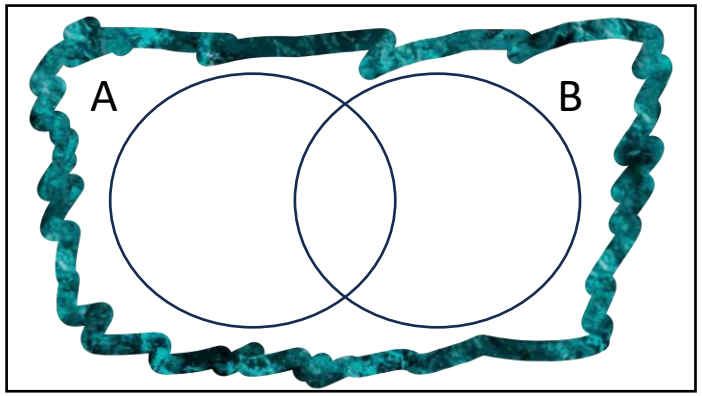
$= \mathbf{\frac{1}{4}}$

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$\frac{3}{4} = \frac{1}{3} + \frac{1}{2} - P(A \cap B)$

$P(A \cap B) = \frac{1}{12}$

# Example 28




*A* and *B* are two events with $P(A) = \frac{1}{3}, P(B) = \frac{1}{2}$ and $P(A \cup B) = \frac{3}{4}$ ,find:

a) $P(B|A)$

b) $P(A'|B')$

c) $P(A'|B)$

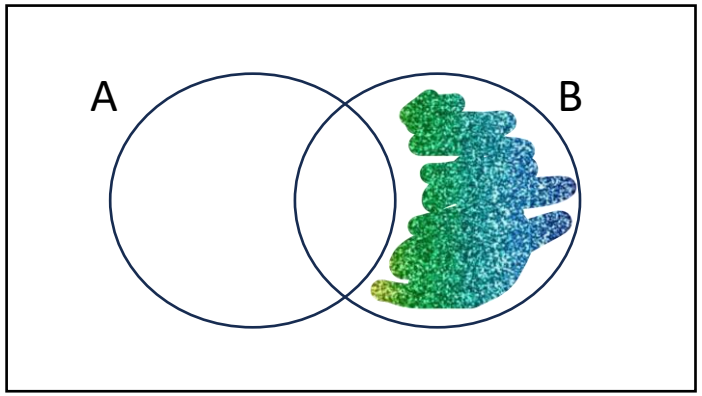
De Morgan's Law

$$P(A' \cap B') = 1 - P(A \cup B)$$
$$= 1 - \frac{3}{4}$$
$$= \frac{1}{4}$$

$$P(B') = 1 - P(B)$$
$$= 1 - \frac{1}{2}$$
$$= \frac{1}{2}$$

b) $$P(A'|B') = \frac{P(A' \cap B')}{P(B')}$$
$$= \frac{1/4}{1/2}$$
$$= \mathbf{\frac{1}{2}}$$

# Example 28




*A* and *B* are two events with $P(A) = \frac{1}{3}, P(B) = \frac{1}{2}$ and $P(A \cup B) = \frac{3}{4}$ ,find:

a) $P(B|A)$

b) $P(A'|B')$

c) $P(A'|B)$

c) $\mathrm{P}(A'|\mathrm{B}) = \frac{\mathrm{P}(A' \cap \mathrm{B})}{\mathrm{P}(\mathrm{B})}$

$= \frac{{}^{5}/_{12}}{{}^{1}/_{2}}$

$= \mathbf{\frac{5}{6}}$

$P(A' \cap B) = P(A \cup B) - P(A)$

$= \frac{3}{4} - \frac{1}{3}$

$= \frac{5}{12}$

# Example 29

A certain company asked all its 100 employees whether they agree or disagree with pay cuts in the present state of the economic slowdown. The table below gives the responses of the survey:

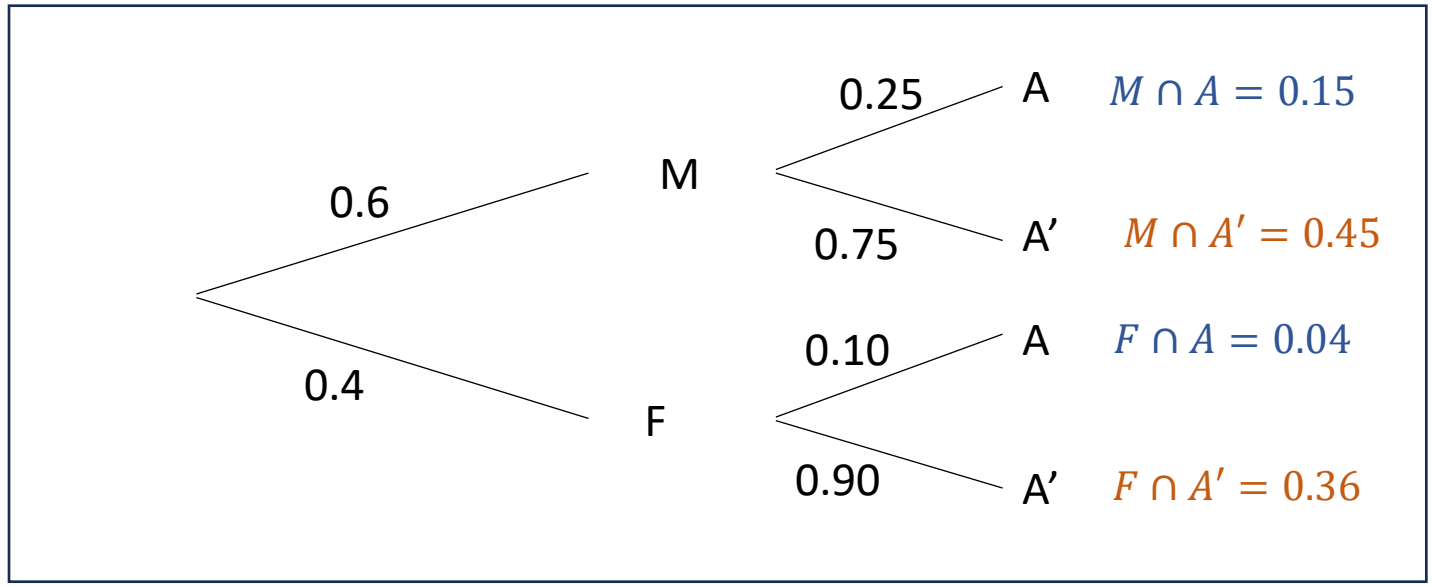
| | Agree | Disagree |
|---|---|---|
| Male | 15 | 45 |
| Female | 4 | 36 |

a) Find the probability that a male employee agrees to pay the cut.

b) Find the probability that a randomly selected employee is a female given that this employee is NOT in favor of the pay cut.

c) Find P(Male|Agree).

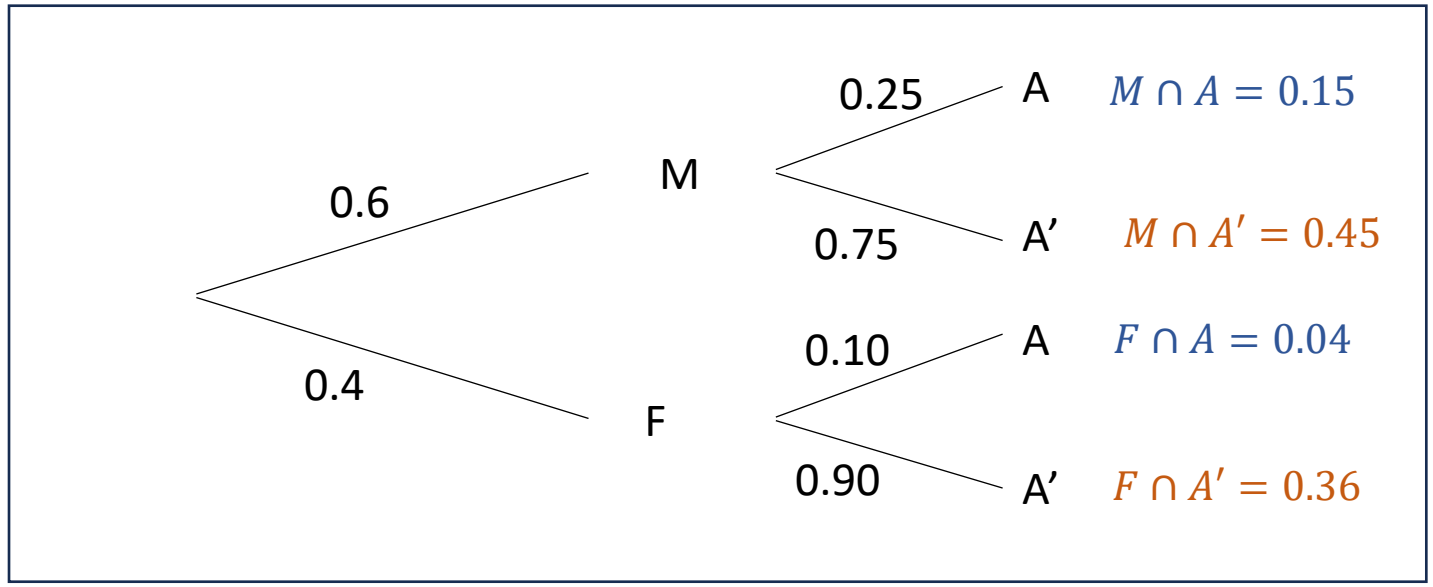
# Example 29




a) $\mathrm{P}(\mathrm{M} \cap A) = 0.6 \times 0.25$

$= \mathbf{0.15}$

# Example 29



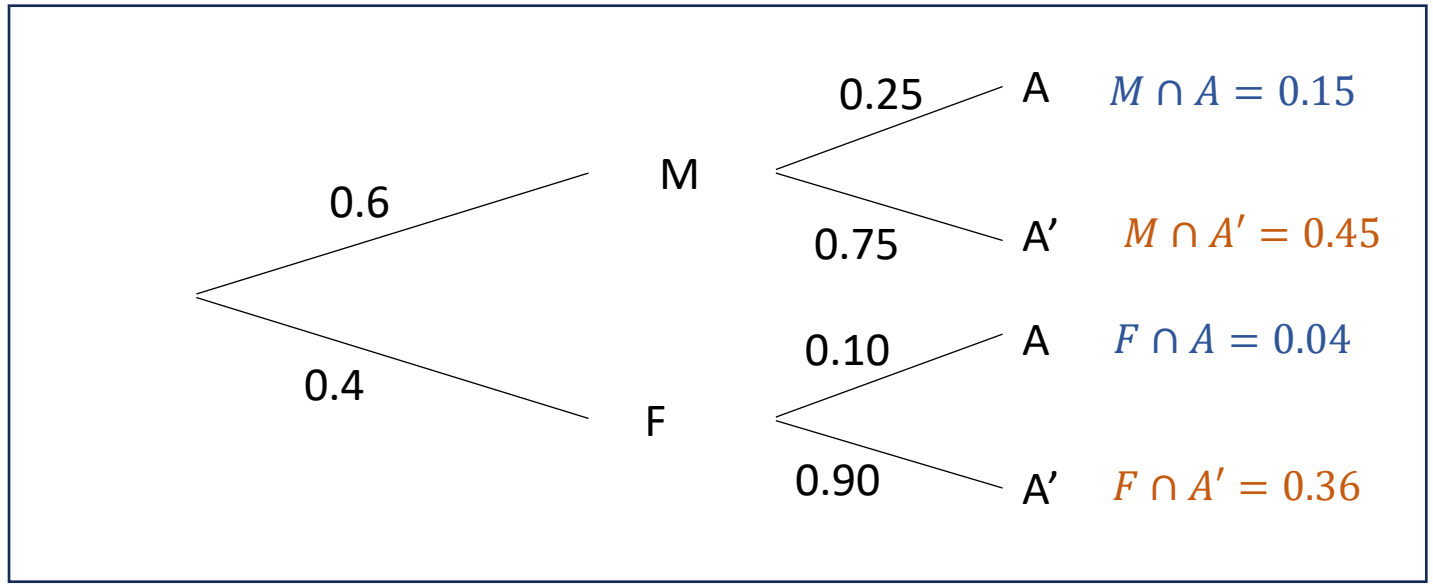

b) $\mathrm{P(F|A')} = \dfrac{\mathrm{P(F \cap A')}}{\mathrm{P}(A')}$

$= \dfrac{0.36}{0.81}$

$= \mathbf{0.4444}$

$P(A') = P(M \cap A') - P(F \cap A')$
$= 0.45 + 0.36$
$= 0.81$

# Example 29




c) $\mathrm{P(M|A)} = \dfrac{\mathrm{P}(M \cap A)}{\mathrm{P}(A)}$

$= \dfrac{0.15}{0.19}$

$= \mathbf{0.7895}$

$P(A) = P(M \cap A) + P(F \cap A)$
$= 0.15 + 0.04$
$= 0.19$

# Independent Events

**Definition 7**:

Two events *A* and *B* are independent if and only if

$$P(B|A) = P(B) \quad \text{or} \quad P(A|B) = P(A)$$

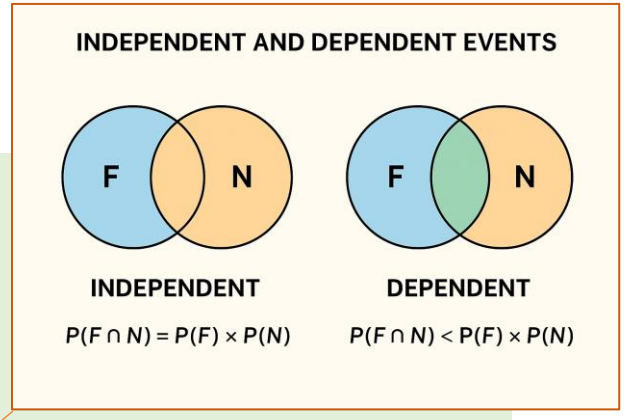
Assuming the existences of the conditional probabilities. Otherwise, *A* and *B* are dependent.

# Example 30

The events F and N are such that P(F) = 0.64, P(N) = 0.33 and P(F or N) = 0.84. Find the probability of:

a) Not F and Not N.

b) Are events F and N independent? State your reason.




a) Not F and Not N = complement of F ∪ N

$$\begin{aligned} P(F' \cap N') &= 1 - P(F \cup N) \\ &= 1 - 0.84 \\ &= 0.16 \end{aligned}$$

b)

| STEP 1: | STEP 2: |
| --- | --- |
| $P(F \cup N) = P(F) + P(N) - P(F \cap N)$<br>$0.84 = 0.64 + 0.33 - P(F \cap N)$<br>$P(F \cap N) = \mathbf{0.13}$ | $P(F) \times P(N) = 0.64 \times 0.33$<br>$= 0.2112 \neq 0.13$<br>**No, they are not independent (one affects the likelihood of the other)** |

# Product Rules or Multiplicative Rules

Enable to calculate the probability that two events will both occur.

**Theorem 12:**

If an experiment the events *A* and *B* can both occur, then

$$P(A \cap B) = P(A) \cdot P(B|A) \quad provided\ P(A) > 0$$

# Example 31

The employees of an accounting firm were classified according to their levels of education and gender, as summarized in the table below:

| **Gender** | **Level of Education** | | | |
|---|---|---|---|---|
| | **Certificate** | **Diploma** | **Degree** | **Master** |
| **Male** | 5 | 7 | 2 | 1 |
| **Female** | 8 | 10 | 5 | 2 |

a) Find the probability that a randomly selected employee is a female with diploma.

b) It is known that the employee is a male, find the probability that he has only a certificate.

# Example 31

| Gender | Level of Education | | | | Total |
|---|---|---|---|---|---|
| | Certificate (C) | Diploma (DIP) | Degree (DEG) | Master (MS) | |
| **Male (M)** | 5 | 7 | 2 | 1 | **15** |
| **Female (F)** | 8 | 10 | 5 | 2 | **25** |
| **Total** | **13** | **17** | **7** | **3** | **40** |

a) $P(F \cap D) = \frac{n(F \cap D)}{n(S)}$

$= \frac{10}{40}$

$= \mathbf{0.25}$

b) $P(C|M) = \frac{P(C \cap M)}{P(M)}$

$= \frac{5/40}{15/40}$

$= \mathbf{0.3333}$

# Product Rules or Multiplicative Rules

**Theorem 13:**

Two events *A* and *B* are independent. If and only if

$$P(A \cap B) = P(A) \cdot P(B)$$

# Example 32

A coin is tossed 3 times

A: event that head occurs on $1^{st}$ two toss – {HHH, HHT}

B: event that tail occurs in the $3^{rd}$ toss – {HHT, HTT, THT, TTT}

C: event that two tail occur in the 3 toss – {HTT, TTH, THT}

Show that A and B are independent, B and C are dependent.

# Example 32

| Events | ( P(A) ) | ( P(B) ) | ( P(C) ) | ( P(\text{Intersection}) ) | Relation |
|---|---|---|---|---|---|
| A & B | 1/4 | 1/2 | — | 1/8 | **Independent** |
| B & C | — | 1/2 | 3/8 | 1/4 | **Dependent** |

Toss a coin 3 times:

$$S = \{HHH, HHT, HTH, THH, TTH, TTT, HTT, THT\}$$

$$n(A) = 2 \quad ; \quad P(A) = \frac{2}{8} = \frac{1}{4}$$

$$n(B) = 4 \quad ; \quad P(B) = \frac{4}{8} = \frac{1}{2}$$

$$n(C) = 3 \quad ; \quad P(C) = \frac{3}{8}$$

a) $P(A) \times P(B) = P(A \cap B)$

$$\frac{1}{4} \times \frac{1}{2} = \frac{1}{8}$$

$$\frac{1}{8} = \frac{1}{8}$$

$A \cap B = \{HHT\}$ $-$ *common element*

$$P(A \cap B) = \frac{1}{8}$$

$\therefore$ ***yes A and B independent***

# Example 32

| Events | ( P(A) ) | ( P(B) ) | ( P(C) ) | ( P(\text{Intersection}) ) | Relation |
|---|---|---|---|---|---|
| A & B | 1/4 | 1/2 | — | 1/8 | **Independent** |
| B & C | — | 1/2 | 3/8 | 1/4 | **Dependent** |

Toss a coin 3 times:

$$S = \{HHH, HHT, HTH, THH, TTH, TTT, HTT, THT\}$$

$$n(A) = 2 \quad ; \quad P(A) = \frac{2}{8} = \frac{1}{4}$$

$$n(B) = 4 \quad ; \quad P(B) = \frac{4}{8} = \frac{1}{2}$$

$$n(C) = 3 \quad ; \quad P(C) = \frac{3}{8}$$

b) $P(B) \times P(C) = P(B \cap C)$

$$\frac{1}{2} \times \frac{3}{8} = \frac{2}{8}$$

$$\frac{3}{16} \neq \frac{2}{8}$$

$B \cap C = \{HTT, THT\}$ $-2$ *common element*

$P(B \cap C) = \frac{2}{8}$

$\therefore$ ***yes B and C are dependent***

# Example 32

| Events | P(A) | P(B) | P(C) | P (Intersection) | Relation |
|---|---|---|---|---|---|
| A & B | 1/4 | 1/2 | — | 1/8 | **Independent** |
| B & C | — | 1/2 | 3/8 | 1/4 | **Dependent** |

# Example 33

Events M and N are such that P(M) = 0.375, P(N) = 0.25, and P(M∪N) = 0.5. Find:

a) $P(M \cap N)$

b) $P(M \cap N')$

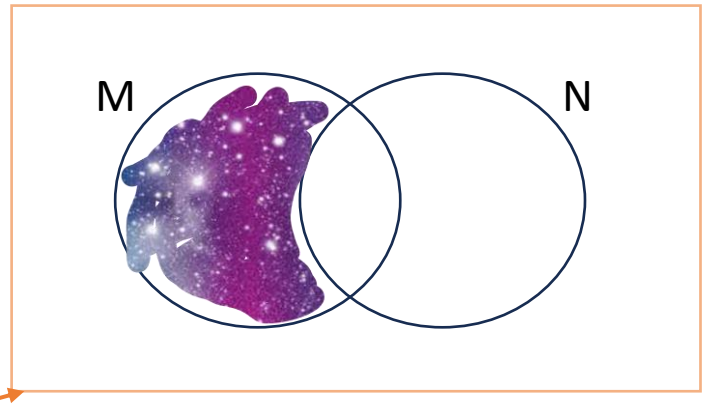
c) Are M and N independent events? Give your reason.

a) $P(M \cup N) = P(M) + P(N) - P(M \cap N)$

$0.5 = 0.375 + 0.25 - P(M \cap N)$

$P(M \cap N) = \mathbf{0.125}$

# Example 33




Events M and N are such that P(M) = 0.375, P(N) = 0.25, and P(M∪N) = 0.5. Find:

a) $P(M \cap N)$

b) $P(M \cap N')$

c) Are M and N independent events? Give your reason.

b) $P(M \cap N') = P(M) - P(M \cap N)$

$= 0.375 - 0.125$

$= \mathbf{0.25}$

c) $P(M) \times P(N) = P(M \cap N)$

$0.375 \times 0.25 = 0.125$

$0.094 \neq 0.125$

$\therefore$ ***M and N are dependent***

# Bayes' Rule

Theorem 14: If the events $B_1, B_2, \ldots . B_k$ constitute a partition of the sample space *S* such that $P(B_i) \neq 0$ for $i = 1,2, \ldots , k$ then for any event A of *S*

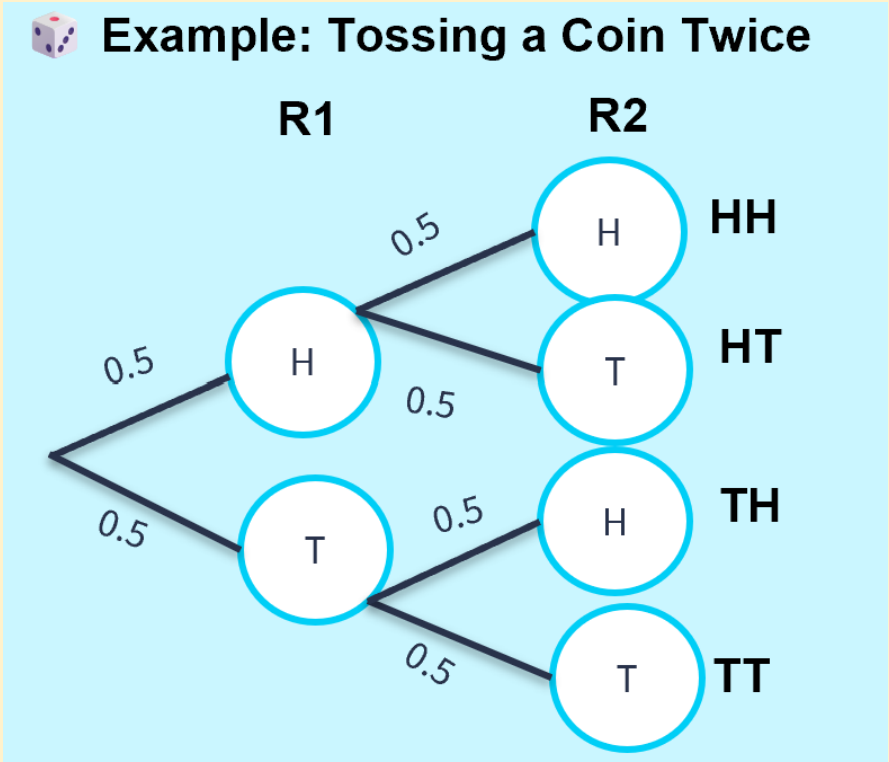
$$P(A) = \sum_{i=1}^{k} P(B_i \cap A) = \sum_{i=1}^{k} P(B_i) \cdot P(A|B_i)$$

$$= P(B_1) \cdot P(A|B_1) + P(B_2) \cdot P(A|B_2) \ldots$$

A tree diagram is used to display the outcomes of an experiment which consist of a series of activities. This ensure that all logical possibilities are considered.

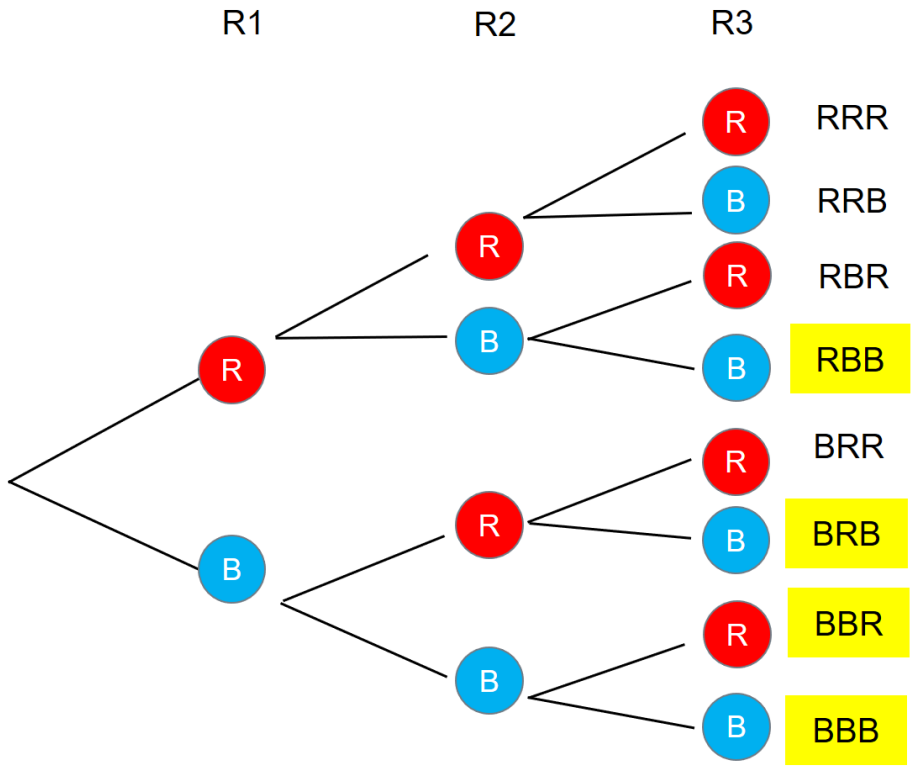
# Tree Diagram

- A **tree diagram** is a visual tool used to list all possible outcomes of a **probability experiment** or **sequence of events** in a structured, branching format.
- **Why Use a Tree Diagram?**
  - To break down **compound or multi-stage events**
  - To **organize outcomes** clearly
  - To help calculate **probabilities** for different paths
  - To identify **all possible combinations** in a logical order
- **Parts of a Tree Diagram:**
  - **Branches** – represent **choices or outcomes** at each stage
  - **Nodes** – the **points** where a choice splits into outcomes
  - **Paths** – from the first event to the final outcome (a **sequence**)



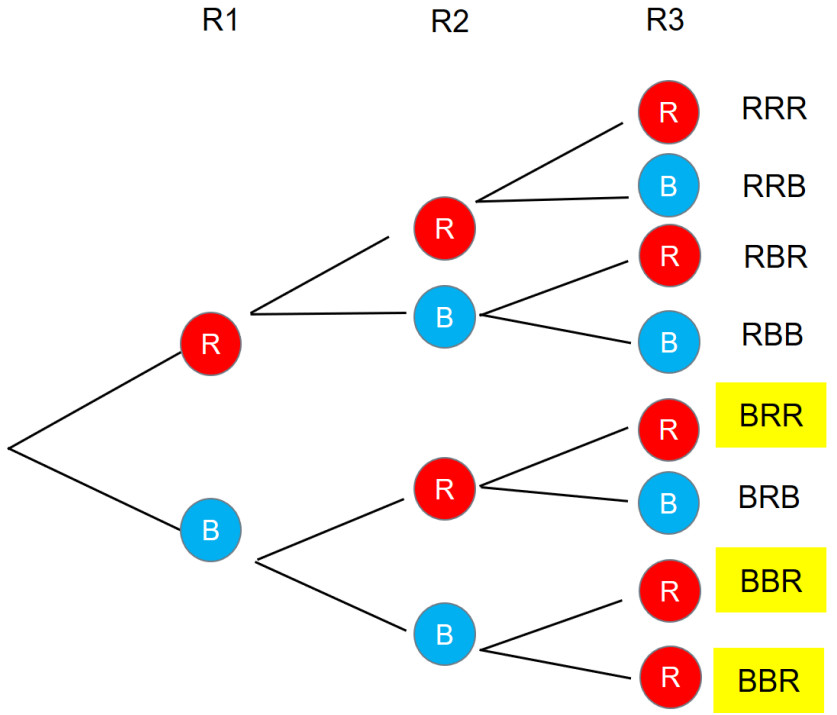
A bag contains of **five red balls** and **two blue balls**. **Three balls** are drawn one by one,

a) with replacement. What is outcome of at least two blue balls selected?




$$P(\leq 2B) = \frac{4}{8} = \frac{1}{2}$$

b) without replacement. What is outcome of getting the first ball blue and the third ball red

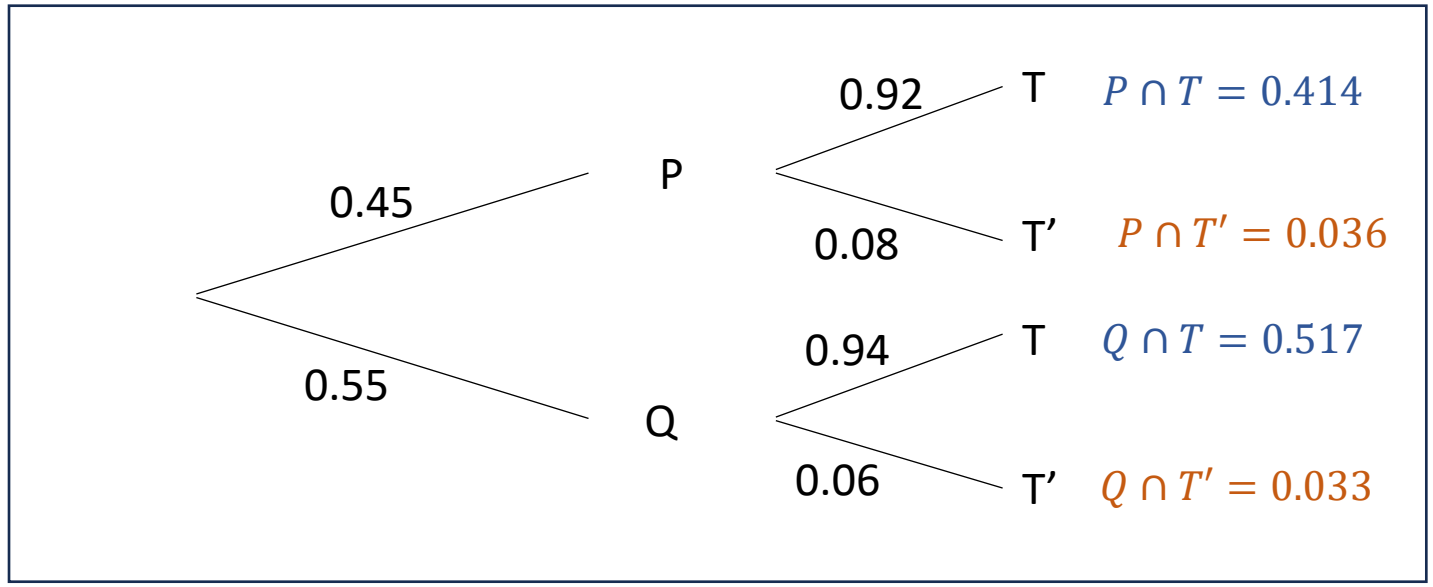



$$P(B_R) = \frac{3}{8}$$

# Example 34

A consulting firm rent cars from two agencies, 45% from agency P and 55% from agency Q. It is known that 8% of the cars from agency P and 6% of the cars from agency Q have bad tires. If a car rented by the firms has good tires, find the probability that it came from agency Q.

# Example 34




a) $P(Q|T) = \frac{P(Q \cap T)}{P(T)}$

$= \frac{0.517}{0.414+0.517}$

$= \mathbf{0.5553}$

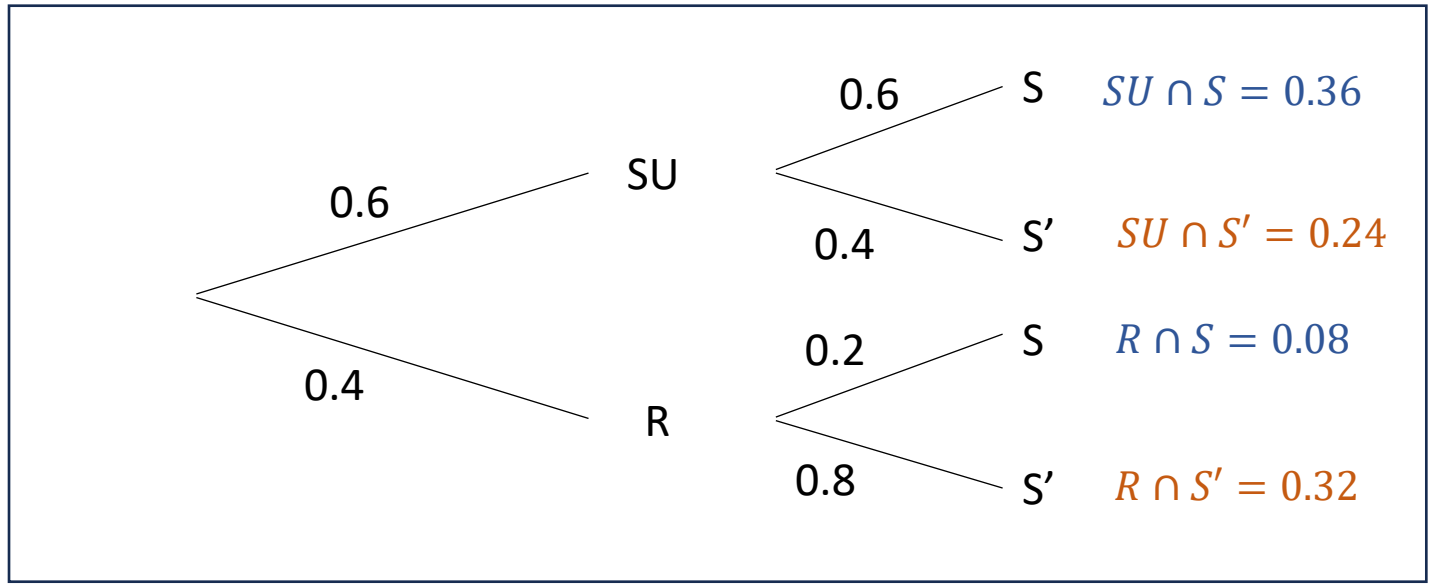
# Example 35

The probability that a particular day will be sunny is 0.6. The probability that Aisyah will be going for a shopping on a sunny day is 0.6 and the probability that she will be going for a shopping on a rainy day is 0.2.

a) Draw a tree diagram.
b) Find the probability that Aisyah will be going for a shopping on a particular day.
c) Find the probability either Aisyah not going for a shopping or it is a sunny day.
d) If Aisyah goes for a shopping on a particular day, find the probability that it is a rainy day.

# Example 35

a)




b) $\mathrm{P(S)} = 0.36 + 0.08 = \mathbf{0.44}$

c) $\mathrm{P(S' \cup SU)} = \mathrm{P(S')} + \mathrm{P(SU)} - \mathrm{P(S' \cap SU)}$

$= (0.24 + 0.32) + 0.6 - 0.24$

$= \mathbf{0.92}$

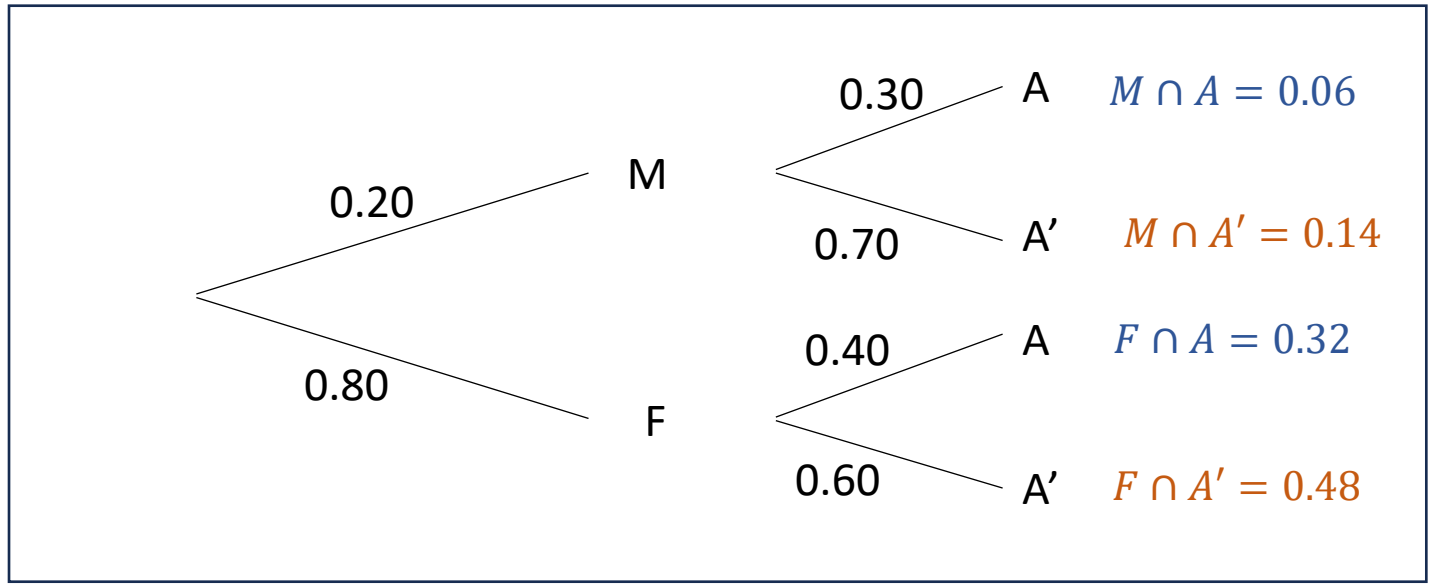
# Example 36

A lecturer noticed that 20% of his students are male, while the rest are female. The probability that a male student obtained grade A in a final exam is 30% while the probability that a female student obtained grade A is 40%.

a) Illustrate the above problem using a tree diagram.
b) Compute the probability that a particular student obtained grade A.
c) If a student failed to obtain grade A, what is the probability that the student was a female?

# Example 36

a)




b) $\mathrm{P(A)} = 0.06 + 0.32 = \mathbf{0.38}$

a) $\mathrm{P(F|A')} = \frac{\mathrm{P}(F \cap A')}{\mathrm{P}(\mathrm{A}')}$

$$= \frac{0.48}{0.48 + 0.14}$$

$$= \mathbf{0.7742}$$

# Past Year Questions (Exercises)

**a) JULY 2025**

AIM Technologies develops two types of data storage devices; 35% are solid-state drives (SSD) and 65% are hard disk drives (HDD). A randomly selected device undergoes a diagnostic check to detect any faults. It is known that 4% of the SSD and 6% of the HDD are found to have faults.

i) Draw a tree diagram to represent the above information.
ii) If the storage device is found to be not faulty during the diagnostic check, find the probability that it is an HDD.

b) **MARCH 2025**

There is an 80% chance that Hamid goes to work by commuter and there is also a 20% chance that he drives his car. If he drives his car to work, the probability that he will be late is 0.25 and if he goes to work by commuter, the probability that he will not be late is 0.88.

i) Find the probability of Hamid not being late to work.
ii) Given that Hamid is late on a particular day, find the probability that he drives his car to work.

# Past Year Questions (Exercises)

**a) FEB 2025**

Rani designs jewelry using beads. She either uses one colour beads to make a piece of jewelry, or mixes beads of different colours. In her current collection, 46% of jewelries are bracelets, 18% are anklets, while the rest are necklaces. Among the bracelets and the necklaces, 48% and 42% are one-coloured respectively. In contrast, 19% of the anklets are multi-coloured. Suppose that a piece of Rani's jewelry collection is picked at random.

a) Find the probability that it is a multi-coloured jewelry.
b) Given that it is a one-coloured jewelry, find the probability that it is an anklet.